

Prompt Engineering Mini-Assessment

Name	A.Siva Krishna
Register No	99230040258
Module	Prompt Engineering
Assessment	Mini Assessment: Multi-Step Prompt Workflow
Course	ICI Gen AI Mastery

1. Assessment Requirements

The mini-assessment requires a 3-step workflow: Step 1 uses few-shot prompting to extract entities, Step 2 uses CoT reasoning to determine urgency, and Step 3 uses JSON mode to produce the final structured result.

- Step 1 — Few-shot: Extract key entities from a support ticket.
- Step 2 — CoT: Reason about the urgency level (low, medium, or high) using Step 1's actual output.
- Step 3 — JSON mode: Output exactly {"entities": [...], "urgency": str, "summary": str}.
- Run the workflow end-to-end on at least one real example and show the intermediate outputs.

2. Workflow Design

SUPPORT TICKET

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STEP 1 — Few-shot Entity Extraction

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Extracted entities

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STEP 2 — CoT Urgency Reasoning

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Urgency: low / medium / high

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STEP 3 — JSON Mode Formatting

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Final JSON: entities + urgency + summary

3. Step 1 — Few-Shot Entity Extraction

Technique: Few-shot prompting

Purpose: Extract the key entities from a support ticket in a consistent format.

You are a support-ticket information extraction assistant.

Extract the key entities from the support ticket.

Return the entities as a bullet list containing only important facts such as:

- Product or service
- Problem or issue
- Customer impact

- Error message
- Account/order information
- Time or deadline

Examples:

Support ticket: "My payment was deducted twice for the same order."

Entities:

- Issue: duplicate payment
- Impact: customer charged twice
- Order: payment for an order

Support ticket: "I cannot log into my account because the password reset email never arrives."

Entities:

- Issue: login failure
- Cause: password reset email not received
- Impact: customer cannot access account

Support ticket: "The application crashes whenever I try to upload a file."

Entities:

- Issue: application crash
- Trigger: file upload
- Impact: customer cannot upload files

Now extract the key entities from this support ticket:

{support_ticket}

4. Step 2 — CoT Urgency Reasoning

Technique: Chain-of-Thought (CoT) reasoning

Purpose: Determine urgency from the actual entities produced by Step 1.

You are a senior customer-support triage specialist.

Based ONLY on the entities extracted in Step 1, determine the urgency level of the support ticket.

Urgency levels:

- **LOW:** Minor issue with little immediate impact.
- **MEDIUM:** Important issue that affects the customer's normal use but is not critical.
- **HIGH:** Critical issue involving major service disruption, financial impact, security, or inability to access an essential service.

Think through the evidence step by step:

1. Identify the main problem.
2. Identify the customer impact.
3. Determine how severe or time-sensitive the impact is.
4. Assign LOW, MEDIUM, or HIGH urgency.

Entities extracted from Step 1:

{step1_output}

Final urgency: LOW, MEDIUM, or HIGH.

5. Step 3 — JSON Mode Formatting

Technique: JSON mode / structured output

Purpose: Combine the intermediate results into one valid, parseable JSON object.

You are a support-ticket response formatter.

Using the extracted entities and urgency from the previous steps, produce **ONLY** a valid JSON object.

The JSON must contain exactly these three keys:

```
{
  "entities": [],
  "urgency": "",
  "summary": ""
}
```

Rules:

- "entities" must contain the key entities extracted in Step 1.
- "urgency" must contain exactly one of: "low", "medium", "high".
- "summary" must be a concise one-sentence summary of the support ticket.
- Return valid JSON only.
- Do not include Markdown.
- Do not include explanations outside the JSON.

Entities:

{step1_output}

Urgency:

{step2_output}

Support ticket:

{support_ticket}

6. End-to-End Test

Test support ticket: “My account has been locked after several failed login attempts, and I cannot access my dashboard. I need access urgently because I have an important report to submit today.”

Step 1 Output

- Issue: account locked
- Cause: several failed login attempts
- Impact: cannot access dashboard
- Deadline: important report due today

Step 2 Output

Final urgency: HIGH

Step 3 Final JSON

```
{
  "entities": [
    "Issue: account locked",
    "Cause: several failed login attempts",
    "Impact: cannot access dashboard",
    "Deadline: important report due today"
  ],
```

```
"urgency": "high",  
"summary": "The customer's account is locked and prevents access to the dashboard before an important report deadline today."  
}
```

7. Rubric Check

Requirement	How the solution satisfies it
Correct technique at each step	Step 1 = few-shot; Step 2 = CoT; Step 3 = JSON mode.
Step 2 uses Step 1 output	The {step1_output} placeholder passes the actual intermediate output into Step 2.
Valid final JSON	Step 3 requires JSON only and uses the exact entities, urgency, and summary keys.
End-to-end test	A complete support-ticket example is included with Step 1, Step 2, and Step 3 outputs.

8. Submission Checklist

- Create the three steps in the assessment UI.
- Paste the Step 1 few-shot prompt.
- Connect Step 1 output to Step 2.
- Paste the Step 2 CoT prompt.
- Connect Step 1 output and Step 2 output to Step 3.
- Enable JSON mode / structured JSON output for Step 3.
- Run the complete workflow on at least one support ticket.
- Show the intermediate outputs and final JSON when submitting.